

Department of Computer Science

Class Test I

Session: 2025-27

Semester: II

Programme: M.Sc. (Computer Science/ Artificial Intelligence)

Course code: CSC82DC00904/CAI82DC01104

Course title: Research Methodology

Max Points: 10

Duration: 1hour

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- Question 1: What is relevance of scientific method in research? [2]
Question 2: Discuss basic principles of experimental design. [3]
Question 3 Discuss LS-Design. What are advantages of LS Design over RB design? [3+2]

Department of Computer Science

Class Test II

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Programme: M.Sc. (Computer Science/ Artificial Intelligence)

Course code: CSC82DC00904/CAI82DC01104

Course title: Research Methodology

Max Points: 10

Duration: 1hour

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- Question 1: Explain measures of dispersion along with respective statistical formula. [3]
Question 2: A random sample of 100 people shows that 25 have opened IRA (Individual retirement arrangement) this year. Construct a 95% confidence interval for the true proportion of people who opened IRA. [4]
Question 3 (a) State the techniques of interpretation.
(b) List the different steps in writing the report. [1+2]



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CENTRAL UNIVERSITY OF SOUTH BIHAR

Question Paper of End-Term Examination, May 2026

Name of Department : Department of Computer Science Programme : M.Sc. (CS)
Session : 2025-2027 Semester : II
Course Title : Research Methodology Course Code : CSC82DC00904
Course Credit : 4 Max. Marks : 70 Duration : 3 hours
Enrolment No. (to be filled by Student) :

Instructions: This question paper have three sections. The use of scientific calculator is permitted in this examination.

Section A

This section has ten questions of one point each. All questions are compulsory.

[1×10=10]

- Q1. What do you mean by term *exploratory* research?
- Q2. What are the steps in research process that should be completed before starting the development of the *Research Design*?
- Q3. Explain the term *coding* in data preparation process.
- Q4. What do you understand by the term *systemic bias in sampling*?
- Q5. Write the meaning of *extraneous variables*.
- Q6. Write the name of the statistical tools used in *Measures of dispersion*.
- Q7. Define the term *Modeling*.
- Q8. Compute the *Harmonic Mean* of the numbers 4, 5 and 10.
- Q9. What is purpose of footnotes in report writing?
- Q10. What is a *popular report*?

Section B

This section has six questions of six points each. Attempt any five questions.

[5×6=30]

Q1. Discuss the following

[3+3]

- I. Positive Skewness and Negative Skewness of the data set
- II. Coefficient of Skewness and its application

Q2. Ten competitors in a beauty contest are ranked by three judges in the following order:

I st judge	1	6	5	10	3	2	4	9	7	8
II nd judge	3	5	8	4	7	10	2	1	6	9
III rd judge	6	4	9	8	1	2	3	10	5	7

Use the Spearman's coefficient of rank correlation to determine which pair of judges has the nearest approach to common tastes in beauty.

Q3. Discuss the merits and demerits of interview method.

Q4. From a random sample of 36 New Delhi civil service personal, the mean age and the sample standard deviation were found to be 40 years and 4.5 years respectively. Construct a 95 percent confidence interval for the mean age of civil servant in New Delhi.

[The standard variate, z , for 95 percent confidence is 1.96, as per the normal curve area table]

Q5. Discuss the modeling process with a neat diagram.

Q6. Differentiate the following

[3+3]

- I. Descriptive vs. Analytical research
- II. Applied vs. Fundamental research

Section C

This section has three questions of fifteen points each. Attempt any two questions.

[15×2=30]

- Q1. (a) Discuss the Hypothesis testing for variance.
(b) Solve the following question using χ^2 distribution.

A pharmaceutical company is considering the purchase of new bottling machines to increase efficiency. The factory currently makes use of machines that fill cough syrup bottles whose volume of medicine has a standard deviation of 1.6 mL. The new machine they are considering was tested in 30 bottles, producing a batch with a standard deviation of 1.2 mL. Does this machine produce a standard deviation less than 1.6 mL? Assuming normal distribution, test at 5% significance level. [Critical values tables are attached here for your reference].

[7+8]

- Q2. Discuss the following informal research designs with the help of the neat and clean diagram. Also mention the limitation of each of the designs given below. [7.5+7.5]

- I. Before-and-after without control design
- II. After only with control design

- Q3. Explain the term Interpretation in report writing. Also explain the different steps in writing the report?

[7.5+7.5]

Table 3: Critical Values of χ^2

Degrees of freedom	Probability under H_0 that of $\chi^2 >$ Chi square						
	99	95	50	10	05	02	01
1	.000157	.00393	.455	2.706	3.841	5.412	6.635
2	.0201	.103	1.386	4.605	5.991	7.824	9.210
3	.115	.352	2.366	6.251	7.815	9.837	11.341
4	.297	.711	3.357	7.779	9.488	11.668	13.277
5	.554	1.145	4.351	9.236	11.070	13.388	15.086
6	.872	1.635	5.348	10.645	12.592	15.033	16.812
7	1.239	2.167	6.346	12.017	14.067	16.622	18.475
8	1.646	2.733	7.344	13.362	15.507	18.168	20.090
9	2.088	3.325	8.343	14.684	16.919	19.679	21.666
10	2.558	3.940	9.342	15.987	18.307	21.161	23.209
11	3.053	4.575	10.341	17.275	19.675	22.618	24.725
12	3.571	5.226	11.340	18.549	21.026	24.054	26.217
13	4.107	5.892	12.340	19.812	22.362	25.472	27.688
14	4.660	6.571	13.339	21.064	23.685	26.873	29.141
15	4.229	7.261	14.339	22.307	24.996	28.259	30.578
16	5.812	7.962	15.338	23.542	26.296	29.633	32.000
17	6.408	8.672	16.338	24.769	27.587	30.995	33.409
18	7.015	9.390	17.338	25.989	28.869	32.346	34.805
19	7.633	10.117	18.338	27.204	30.144	33.687	36.191
20	8.260	10.851	19.337	28.412	31.410	35.020	37.566
21	8.897	11.591	20.337	29.615	32.671	36.343	38.932
22	9.542	12.338	21.337	30.813	33.924	37.659	40.289
23	10.196	13.091	22.337	32.007	35.172	38.968	41.638
24	10.856	13.848	23.337	32.196	36.415	40.270	42.980
25	11.524	14.611	24.337	34.382	37.652	41.566	44.314
26	12.198	15.379	25.336	35.363	38.885	41.856	45.642
27	12.879	16.151	26.336	36.741	40.113	44.140	46.963
28	13.565	16.928	27.336	37.916	41.337	45.419	48.278
29	14.256	17.708	28.336	39.087	42.557	46.693	49.588
30	14.953	18.493	29.336	40.256	43.773	47.962	50.892

Note: For degrees of freedom greater than 30, the quantity $2\chi^2 - \sqrt{2d.f. - 1}$ may be used as a normal variate with unit variance i.e., $z_\alpha = \sqrt{2\chi^2} - \sqrt{2d.f. - 1}$.